

Erdős Problem #616

EP-616

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $r \geq 3$. For an r -uniform hypergraph G let $\tau(G)$ denote the covering number (or transversal number), the minimum size of a set of vertices which includes at least one from each edge in G .

Determine the best possible t such that, if G is an r -uniform hypergraph G where every subgraph G' on at most $3r - 3$ vertices has $\tau(G') \leq 1$, we have $\tau(G) \leq t$.

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>